

# Density Effects on the Flow Structure in Multi-fluid Shock-turbulence Interaction

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**Turbulence-resolving shock-capturing simulations of isotropic turbulence interaction with a normal shock wave are conducted for small (single-fluid) and large (multi-fluid) density variations. This is an extension of the canonical Shock-Turbulence Interaction (STI) problem, involving significant variable density effects. Numerical accuracy of simulations has been established using a series of grid and LIA convergence tests. Eulerian and Lagrangian analyses of numerical data reveal several important features of the post-shock turbulence structure. For example, results show that the increase in the post-shock correlation between rotation and strain is weakened in the multi-fluid case. Consistent with the previous single-fluid studies and LIA predictions, the shock wave significantly changes the topology of the turbulent velocity field, exhibiting a symmetrization of the joint PDF of second and third invariant of anisotropic velocity gradient tensor. In the multi-fluid case, this trend is more significant and mainly manifested in the heavy fluid regions. Such symmetrization is shown to be connected to the decreased vortex stretching rate across the shock wave. The residence times in each quadrant and their contributions to vortex stretching during post-shock return-to-isotropy process are studied vis Lagrangian/particle statistics.**

## I. Introduction

Canonical Shock-Turbulence Interaction (STI) problem is an important fundamental problem that has been extensively studied [1–13]. Understanding this problem is a prerequisite for modeling complex phenomena such as inertial confinement fusion (ICF), hypersonic combustion, or astrophysical explosions. Ribner [14] proposed a theoretical model for the description of STI, in which linearized Euler equations and Rankine-Hugoniot equations were solved. In this theory, referred to as Linear Interaction Approximation (LIA), the pre-shock turbulence was assumed to be isotropic and made of small amplitude disturbances, so nonlinear and viscous effects were excluded from the analysis. The early DNS study of shock-isotropic turbulence interaction by [12] was limited to relatively low flow Mach numbers (1.2 or less), and showed reasonably good agreement with LIA when turbulent Mach number  $M_t$  was kept small. Mahesh et al. [8], Jamme et al. [11] have also studied the effects of different types of upstream turbulence on STI using turbulence resolving simulations. Larsson et al. [9], Lee et al. [10], Larsson and Lele [13] have conducted high-order shock-capturing turbulence-resolving simulations for different flow Mach numbers, and found good agreement with LIA predictions for statistics like turbulent kinetic energy and vorticity variance, at relatively low  $M_t$ . More recently, Ryu and Livescu [7] have conducted shock resolving DNS for a wide range of parameters and have shown that DNS results converge to LIA when the ratio of shock thickness  $\delta_s$  to Kolmogorov length scale  $\eta$  becomes small, even at low upstream Reynolds numbers. In Livescu and Ryu [4], Ryu and Livescu [7], the single-fluid STI data generated using shock-resolving DNS and shock-LIA procedure are used to study the topological structure of turbulence. LIA and shock-capturing simulation data have also been used for a detailed study of the turbulent energy flux in the post-shock region [15].

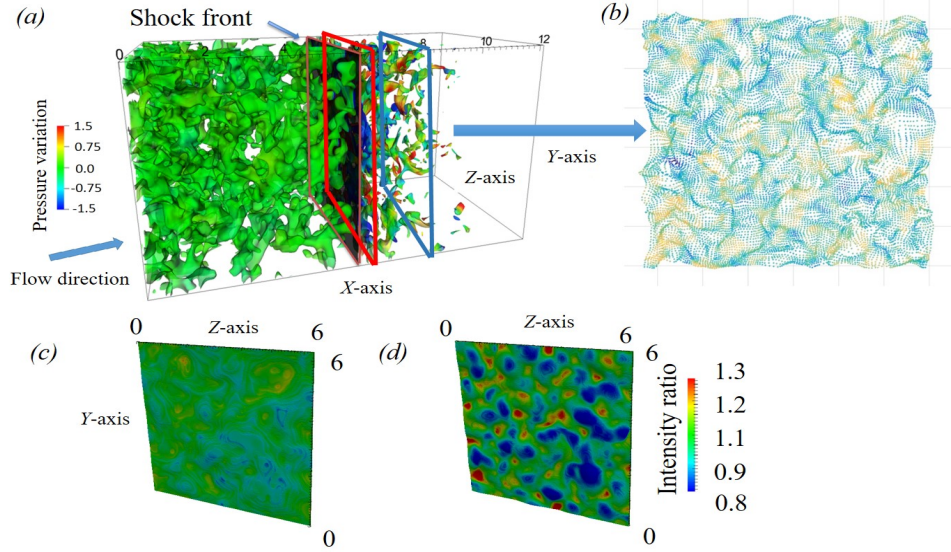
All of the above studies (and most of other relevant works not discussed here) were focused on STI for single-fluid or low-density variation flows. Recently, we have used an accurate shock-capturing method to study the variable density effects on STI by conducting multi-fluid STI simulations [16, 17]. The general features of the multi-fluid STI at relatively low Atwood number of  $A_t = 0.28$  are identified in this work [16, 17]. For example, it is shown that

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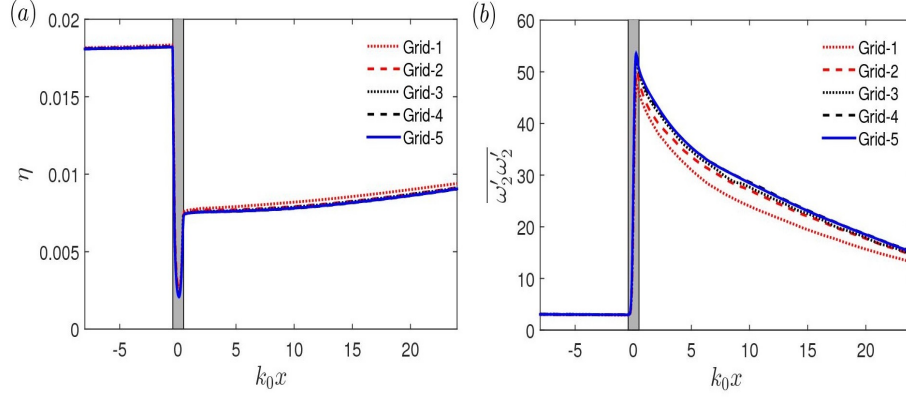
**Fig. 1** Instantaneous contours of 3D pressure fluctuations and shock surface in isotropic turbulence interacting with a Mach 2 shock. (a) The iso-surface of heavy fluid mole fraction ( $\phi = 0.25$ ) is colored by instantaneous pressure fluctuations. The instantaneous shock surface is colored by the local pressure jump across the shock for (b) multi-fluid and (c) single-fluid cases.

multi-fluid STI further enhances the amplification of TKE and the reduction of turbulent length scales. In this paper, we study the density effects on post-shock turbulence structure. Results obtained from both grid/Eulerian data and fluid particle/Lagrangian data concerning the structure of the post-shock turbulence are presented and discussed below.

## II. Computational Setup and Numerical Accuracy

In this study, the conservative form of the dimensionless compressible Navier-Stokes equations for continuity, momentum, energy, and scalar are solved numerically together with the perfect gas law using a high-order hybrid numerical method. The inviscid fluxes for continuity, momentum, and energy are computed by the fifth-order MP scheme, as described in [18]. The scalar equations are also computed in their conservative forms using the MP scheme. The viscous fluxes are calculated by the sixth-order compact scheme [19]. The 3rd-order Runge-Kutta scheme is used for time advancement. Lagrangian statistics are extracted from the turbulent flow by solving the Lagrangian particle equations for non-inertial fluid particles. The cubic spline scheme, whose accuracy in fluid particle tracking has been studied in [20], is used in the pre- and post-shock regions. More than 1.048 million randomly distributed particles are considered to ensure statistical convergence.

The physical domain for the simulations considered in this paper is a box that has a dimension of  $4\pi$  in the streamwise direction (denoted as  $x$ ) and  $(2\pi, 2\pi)$  in the transverse directions (denoted as  $y$  and  $z$ ), as shown in figure 1 (a). A buffer layer is used at the end of the computational domain from  $4\pi$  to  $6\pi$  to prevent reflecting waves. In the transverse directions, periodic boundary conditions are used so that the flow can be assumed to be periodic and homogeneous in these directions. To provide inflow turbulence, a pre-generated isotropic turbulence box is superposed on the uniform mean flow with Mach number = 2.0 and convected into the domain using Taylor's hypothesis. The isotropic turbulence box generated following [21] has  $M_t \approx 0.1$  and  $Re_\lambda \approx 45$ . For this  $M_t$  value, Taylor's hypothesis is appropriate for approximating spatially developing turbulence with a temporally developing turbulence [22]. The variable density (multi-fluid) effects arise from compositional variations, by correlating the density to an isotropic scalar field. The scalar field is generated as a random field following a Gaussian spectrum with a peak at  $k_s = 8.0$  and has double-delta probability density function distribution so that the scalar value initially is either 1.0 or 0.0. The initial scalar field is smoothed by solving a diffusion equation so that the scalar field can be fully resolved by the chosen mesh. The resulting scalar field is then allowed to decay in the fully developed isotropic turbulence setup for one eddy turn over time as a passive scalar. The mass fractions of the two fluids can be extracted from the density field using the infinite



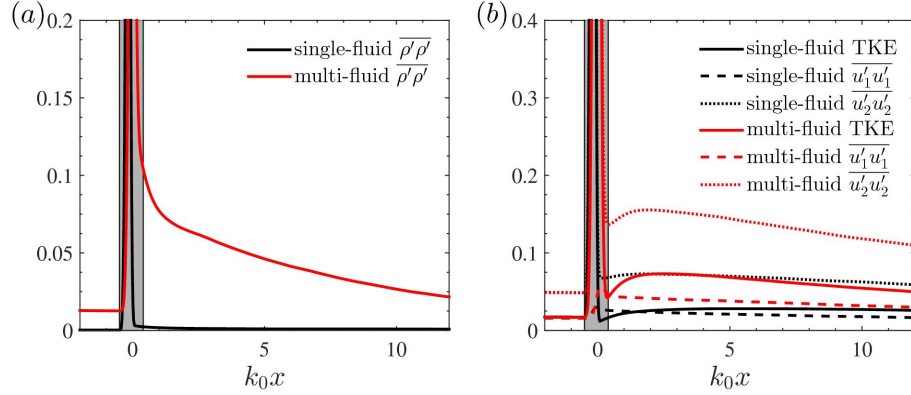
**Fig. 2** Results of multi-fluid grid convergence tests at  $Re_\lambda = 45$  and  $M_t = 0.09$ . Streamwise development of (a) Kolmogorov length scale  $\eta$  and (b) transverse vorticity variance  $\overline{\omega'_2\omega'_2}$  is shown. The region of unsteady shock movement is marked in grey.

speed of sound relation [23, 24], which is approximately satisfied at low turbulent Mach numbers. The Atwood number corresponding to the molar weights of the two fluids is 0.28. This value of the Atwood number was chosen such that the variable density effects are non-negligible, yet the interaction with the shock wave is still in the wrinkled-shock regime. The Prandtl and Schmidt number are the same and equal to 0.75. The mean flow Mach number is 2.0. The turbulent Mach number and the Reynolds number based on the Taylor micro length scale are 0.09 and 44 right before the shock wave.

**Table 1** Details of the mesh used in the grid convergence tests

|        | Mesh info           | $\Delta x_d/\Delta y$ | $(k_{max}\eta)_{min}$ | $\eta/\delta_n$ |
|--------|---------------------|-----------------------|-----------------------|-----------------|
| Grid-1 | $1024 \times 256^2$ | 2.0                   | 0.96                  | 0.93            |
| Grid-2 | $1024 \times 384^2$ | 1.3                   | 1.45                  | 0.93            |
| Grid-3 | $1536 \times 384^2$ | 2.0                   | 1.45                  | 1.40            |
| Grid-4 | $1536 \times 512^2$ | 1.5                   | 1.93                  | 1.40            |
| Grid-5 | $2048 \times 512^2$ | 2.0                   | 1.93                  | 1.86            |

Before discussing our main results, the procedure for obtaining the statistics to achieve statistical convergence is described. Most statistics are spatially averaged over homogeneous (y and z) directions. Time averaging for two pass-over times is also conducted for some of the statistics. All the results are obtained after the flow reaches the statistically steady state, which is achieved after the acoustic wave has propagated from the outlet to the shock wave [9]. The grey region in the following figures indicates the unsteady shock region, inside which the results are affected by the shock wrinkling and unsteady shock movement. To verify the accuracy of our simulations, we have examined the turbulent statistics from two different perspectives. First, grid convergence tests are conducted to ensure that the statistics are grid converged and all turbulence scales are well-resolved. Two factors are considered in assessing the reliability of grid: a) the maximum turbulent wavenumber  $k_{max}$ , multiplied by the Kolmogorov length scale,  $\eta$ , should be considerably greater than one, and b) the scale separation between numerical shock thickness and  $\eta$  should be large enough (numerical shock thickness is calculated using  $(u_{1,u} - u_{1,d})/|\partial u_1/\partial x_1|_{max}$ , where  $\overline{u_{1,u}}$  and  $\overline{u_{1,d}}$  are the pre-shock and post-shock mean streamwise velocities). Based on these two criteria, five different grids are constructed as shown in table 1. Results obtained from simulations using these meshes in figure 2 show that the turbulent statistics, including  $\eta$  and transverse vorticity variance and all other turbulent statistics not shown here, are grid-converged when the finest grid is used with  $k_{max}\eta = 1.93$  and  $\eta/\delta_n = 1.86$ . Increasing either of these two values won't affect the post-shock turbulent statistics. This test proves that the Grid-5 is sufficient for resolving all turbulent length scales.



**Fig. 3** Plots of (a) density variance, (b) TKE for multi-fluid (red) and single-fluid (black) simulations.

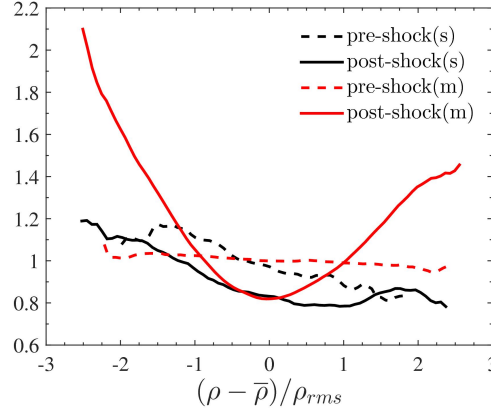
Secondly, LIA convergence tests are conducted following [7] to show that the nonlinear and viscous effects are correctly resolved using current numerical methods. Turbulent Mach number  $M_t$  and Taylor Reynolds number  $Re_\lambda$  are varied in the single-fluid simulations, covering a wide range of parameter space. Results show that shock-capturing simulation results do converge to LIA predictions for individual Reynolds stress components as long as certain conditions are satisfied [17]. For the first time it is shown in [17] that the asymptotic values for Reynolds stresses can be approximated using shock-capturing simulations.

### III. Results and Discussions

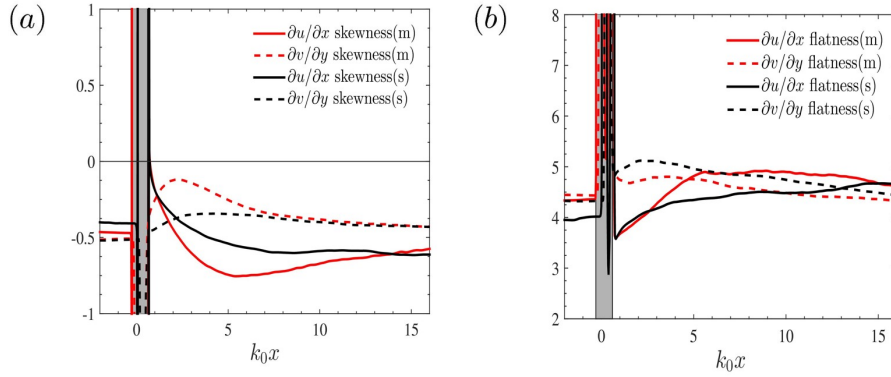
In this section, the results obtained from multi-fluid STI simulation are examined and compared with those of a reference single-fluid case. In this reference case, the mass fraction of the heavy fluid is set to be 1.0, so that the density variations are much smaller compared to the multi-fluid case. Other than the density field, the multi-fluid and the single-fluid simulations are similar.  $Re_\lambda$  and  $M_t$  are chosen to be 45 and 0.09 right upstream of the shock wave based on previous grid convergence and LIA convergence tests results [17].

Several low order post-shock turbulence quantities are discussed first to give a general overview of the multi-fluid STI. Figure 3 shows the density variance, turbulent kinetic energy (TKE), and its components. Compared to the single fluid case, the multi-fluid case features large density variations before the shock wave ( $k_0x < 0.0$ ,  $k_0$  is the wavenumber corresponds to peak energy spectrum), which are mostly due to the mixing of two fluids with different molecular weights. For the single-fluid case, only small density variations exist, which are generated by the turbulence. After the shock wave, the density variations are amplified, and this effect is much more significant in the multi-fluid case. This amplification across the shock is related to the intensified thermodynamic fluctuations. Immediately after the unsteady shock region (marked as grey in figure 3), multi-fluid density variations start to decrease, which is mainly related to the transition from acoustic to vorticity field. After  $k_0 \approx 2.0$ , the decrease in density variance is slower, as the flow becomes dominated by turbulent mixing again. For TKE and its components, the amplification across the shock wave is also much stronger in the multi-fluid case as shown in figure 3 (b), in agreement with our previous findings [16, 17]. The peak TKE position shifts upstream to  $k_0x \approx 2.0$  in contrast to the single-fluid case, where it is at  $k_0x \approx \pi$ . However, this shorter transient length in the multi-fluid case may not be universal, considering that it could be affected by the structure of density field, Atwood number, and other flow parameters.

The conditional expectations of TKE as a function of normalized density are shown in figure 4 to examine the density effects on the preferential deposition of TKE through the interaction. The conditional TKE is also normalized by the peak TKE in the post-shock region. For the multi-fluid case (figure 4 red), we note that TKE is larger in the relatively high or low-density regions after the shock wave. One possible explanation is that in high and low-density regions, the local sound speed has different values from that of average sound speed so that the local shock strength changes significantly. The local shock strength variation further changes the post-shock velocity  $u_{1,d}$  and makes it deviate from the averaged post-shock velocity  $\overline{u_{1,d}}$ . The deviation from  $\overline{u_{1,d}}$  in low and high-density regions is much larger than that in regions with moderate density, which results in larger TKE in the high and low-density regions. The conditional average of  $u_1'$  (not shown) agree with our analysis and support our explanation. We also note that TKE is larger in the light fluid regions compared to heavy fluid regions. This is due to the low inertia of the light fluid, which



**Fig. 4** Conditional expectation of TKE as a function of density at various streamwise locations for single-fluid and multi-fluid simulations



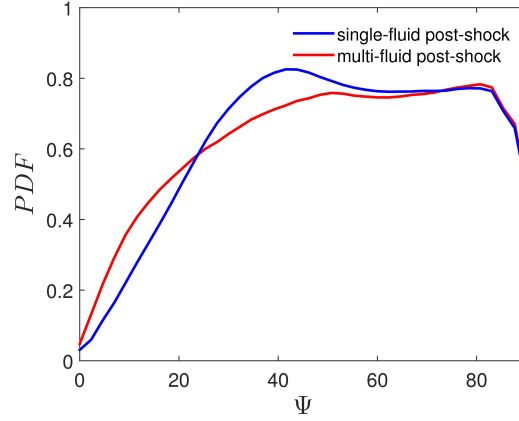
**Fig. 5** Development of streamwise and transverse components of the (a) skewness and (b) flatness tensor

responds faster to the changes in the local strain and, thus, accelerates faster. For the single-fluid case, a similar trend is also observed as shown in figure 4, but much weaker.

#### A. Variable Density Effects on Shock Modification of Turbulent Structure

In our earlier study on multi-fluid STI, we have focused on turbulent length scales, Reynolds stresses, vorticity variance, scalar variance and their budgets [16, 17] as the first quantities to examine and compare with single-fluid STI predictions. In Livescu and Ryu [4], Ryu and Livescu [7], the single-fluid STI data generated using shock-resolving DNS and LIA formulas are used to study the topological structure of turbulence. Here, we aim to extend this analysis to multi-fluid STI.

In figure 5, the development of the skewness and the flatness of the velocity derivatives is examined across the shock wave. For isotropic turbulence, the skewness of the velocity derivatives should be around -0.5, which is observed in the pre-shock region for both single and multi-fluid cases as seen in figure 5 (a). Immediately after the shock wave, the components of the skewness tensor are shown to be modified in very different ways. The streamwise components for both single-fluid and multi-fluid cases become very close to 0.0 and they decrease rapidly as turbulence evolves away from the shock wave. Due to the strong density variations, the multi-fluid case exhibits a faster decrease before  $k_0x = 5.0$ , after which it starts to increase. The shock modification of the transverse component is relatively small, even though both components increase until they reach a peak and then they decrease. Comparably, the multi-fluid case exhibits shorter but more intensified transition. In figure 5 (b), the development of velocity derivative flatness across the shock wave is shown. It is noted that right after the shock wave, the streamwise component of the flatness tensor

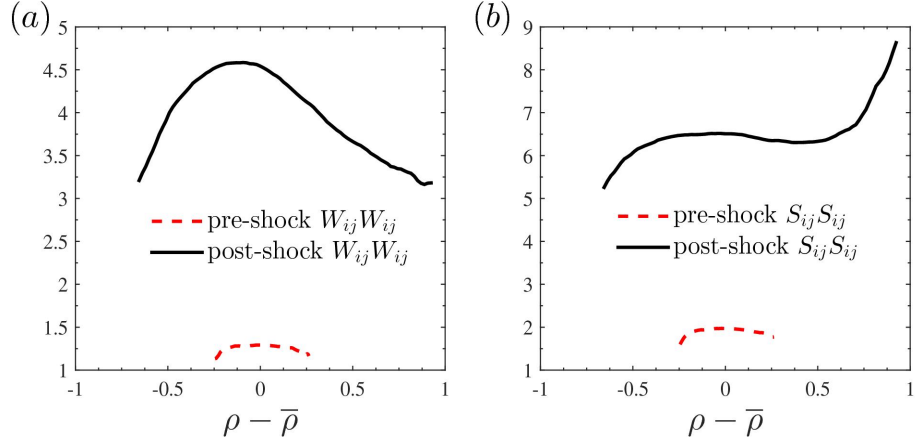


**Fig. 6 PDF of the strain-ensrophy angle  $\Psi$  in degrees for post-shock turbulence.**

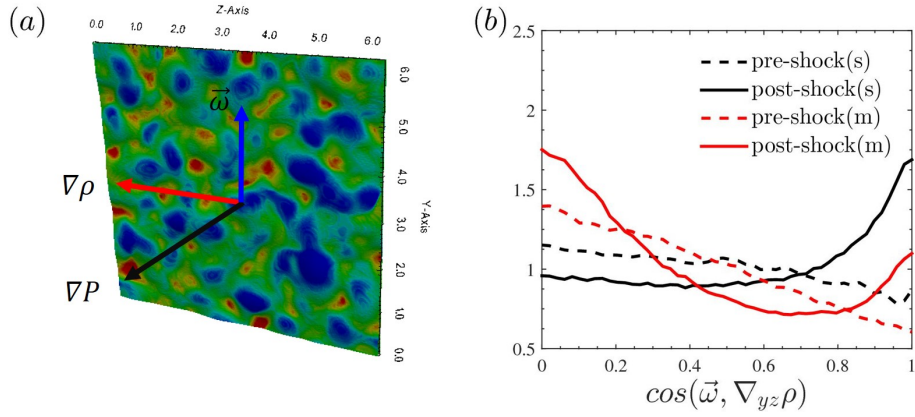
decreases in value while the spanwise component increases. The effect of density variations is relatively small at the Atwood number we considered in this study, so both single-fluid and multi-fluid cases predict similar values. Away from the shock wave, the flatness of streamwise velocity derivative increases, returning to its pre-shock value, while the increase is much faster for the multi-fluid case. For the transverse component, the flatness slowly decreases after a short transient process.

The preferential amplification of the transverse components of the rotation and strain rate tensors is a well-known effect in STI and has been extensively studied for the canonical single-fluid flows [4, 7, 8]. This can lead to an increase in the correlation between the two quantities. When density variations are introduced, Tian et al. [17] showed that the transverse components of vorticity variance are further enhanced across the shock wave while the streamwise component retains almost the same values. To better understand the variable density effects on post-shock turbulence, the PDF of the strain-ensrophy angle,  $\Psi$ , is considered in figure 6.  $\Psi$  is calculated using  $\Psi = \tan^{-1}(S_{ij}S_{ij}/(W_{ij}W_{ij}))$ , where  $S_{ij} = 1/2(A_{ij} + A_{ji})$  and  $W_{ij} = 1/2(A_{ij} - A_{ji})$  are the strain and rotation tensors and  $A = \nabla \vec{v}$  is the velocity gradient tensor. In isotropic turbulence, the PDF of  $\Psi$  peaks at  $90^\circ$  [25], indicating a strain dominated structure. In single-fluid post-shock turbulence, the PDF of  $\Psi$  exhibits a shift of the peak from  $90^\circ$  to around  $45^\circ$ , as the shock Mach number increases. This has been observed by [4] and interpreted as the increase in correlation of strain and rotation. However, in the multi-fluid case, the peak still remains at the large angles and the increase in the correlated region is not as pronounced as that in the single-fluid case, at the same shock Mach number. Figure 6 implies that the rotation and strain are amplified differently by the shock when large density variations are present, which compromises the correlation between these two quantities.

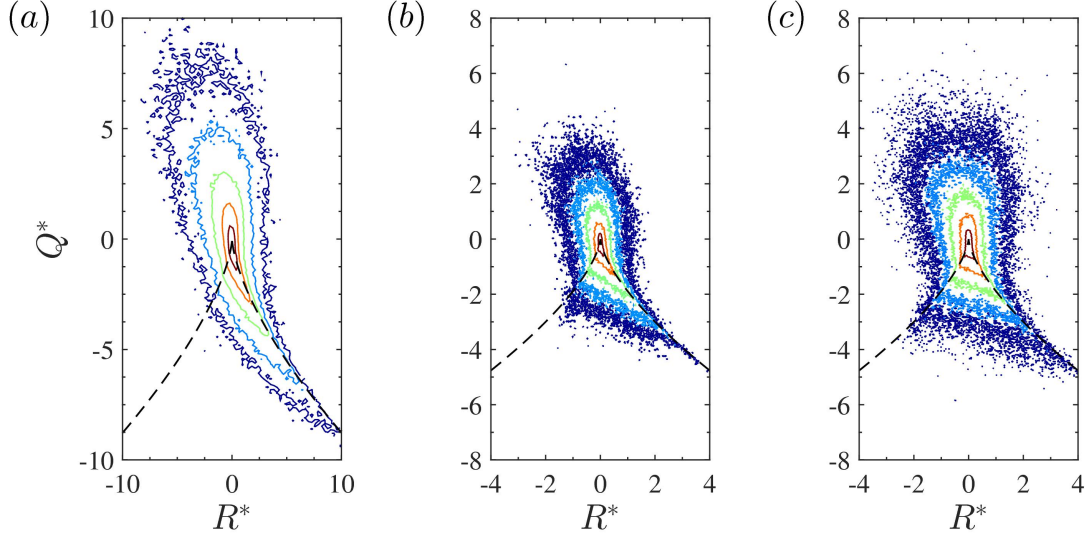
To understand the different roles that density plays through the interaction with the shock wave, the pre-shock and post-shock values of the conditional expectation of vorticity and strain rate tensor, conditioned on density are shown in figure 7 for the multi-fluid case. As observed, the correlation between vorticity (and strain rate) with density is completely different before and after the shock wave. Through the shock wave, the amplification of vorticity is stronger in the mixed fluid region with medium density but weaker in the pure fluid region. This is not observed in the single-fluid simulation. One mechanism that might be responsible for this behavior is the baroclinic torque:  $(\nabla \rho \times \nabla P)/\rho^2$ . Consider a wrinkled shock surface extracted from the multi-fluid simulation as shown in figure 8 (a); a strong pressure gradient  $\nabla P$  exists through the shock wave as marked in the figure; large density gradients  $\nabla \rho$  also exist, especially in the mixed fluid region. Since the pre-shock density field is isotropic, it is possible for  $\nabla \rho$  and  $\nabla P$  to become misaligned, becoming a source of vorticity generation. In the pure fluid region or single-fluid simulation, however, the density gradients are much smaller, making the cross product of  $\nabla P$  and  $\nabla \rho$  also small. To confirm this, the PDF of the angle between density gradient in the transverse direction and vorticity vector is examined in figure 8 (b). Before the shock wave, it is noted that the PDF of the cosine of the angle between these two vector is slightly tilted towards the 0-degree angle. After the shock wave, the multi-fluid case exhibits a stronger tendency of vorticity vector being perpendicular to the density gradient. In contrast, the alignment is completely different after the interaction with the shock wave and they are more likely to be parallel to each other. This provides evidence that density gradients and baroclinic torque play important roles in establishing the preferential deposition of vorticity across the shock wave.



**Fig. 7** Conditional expectation of the magnitude of (a) rotation tensor and (b) strain rate tensor as a function of density at various streamwise locations.



**Fig. 8** (a) Demonstration of the baroclinic torque across the shock wave and (b) PDF of the orientation between vorticity vector and density gradient in y-z direction immediately after the shock wave.



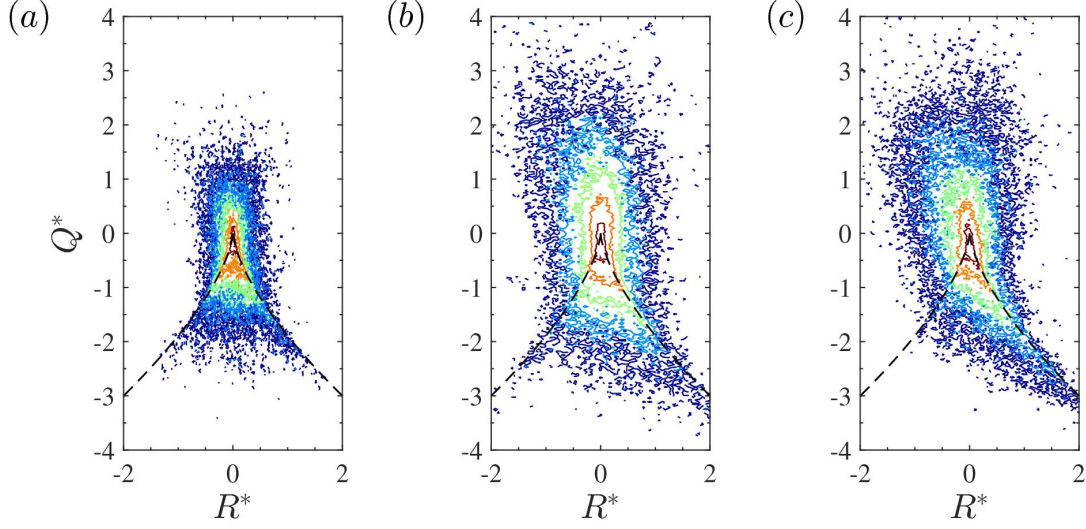
**Fig. 9** Iso-contour lines of joint PDFs of  $(Q^*, R^*)$ , second and third invariant of and anisotropic part of the velocity gradient tensor for (a) pre-shock, (b) single-fluid and (c) multi-fluid cases immediately after the shock. The lateral lines denote the locus of zero discriminant.

On the other hand, figure 7 (b) shows that the magnitude of strain rate tensor tends to be stronger in the heavy fluid region and weaker in the light fluid region. This trend is different from those observed for the vorticity, implying that the roles the density plays through the shock wave are different for vorticity and strain rate tensor, which weakens the move of the strain-entropy angle PDF towards  $45^\circ$ .

To further characterize the turbulence structure behind the shock wave, we have performed an analysis of the invariant plane of the velocity gradient tensor. The second and third invariants (denoted as  $Q$  and  $R$ ) of the anisotropic part of the velocity gradient tensor can reveal important details of turbulence structure. The anisotropic part of the velocity gradient tensor is calculated using the formula  $A^* = A - \theta/3I$ , where  $\theta = \nabla \cdot \vec{v}$ . For isotropic turbulence, the joint PDF has the well-known tear-drop shape which has been observed in boundary layers, mixing layers, and channel flows [26, 27]. This type of distribution of  $Q$  and  $R$  indicates that for these flows, turbulence has a local topology of stable-focus/stretching or unstable-node/saddle/saddle. In figure 9 (a), it is shown that the joint PDF of normalized second and third invariant  $Q^* = Q/(W_{ij}W_{ij})$  and  $R^* = R/(W_{ij}W_{ij})^{3/2}$  for the pre-shock flow has almost the same tear-drop shape. After the shock wave, Ryu and Livescu [7] used shock-LIA and DNS data to show that for single-fluid STI, the shape of the distribution is significantly modified, demonstrating a symmetrization of the joint PDF. This symmetrization indicates that the regions with stable-focus/compression and stable-node/saddle/saddle (first and third quadrant) are enhanced. To understand the variable density effects on this symmetrization, the joint PDFs of  $(Q^*, R^*)$  for both single-fluid and multi-fluid post-shock turbulence are compared in figure 9.

Figure 9 (b) shows the joint PDF iso-contour lines for the single-fluid post-shock turbulence. The dashed lines denote the locus of zero discriminant of  $A^*$ , where  $Q^*$  and  $R^*$  satisfy  $27R^{*2}/4 + Q^{*3} = 0$ . Compared to the pre-shock PDF, there is a tendency towards symmetrization of the invariant distribution, with more points located in the first and third quadrants. This agrees very well with previous shock-LIA results [7]. Similar to single-fluid STI, multi-fluid STI demonstrates a tendency towards symmetrization of the  $(Q^*, R^*)$  distribution. However, the multi-fluid distribution is slightly more symmetric and has larger variance, with more points away from the axes. This implies that more extreme "events" exist in the multi-fluid turbulence. This also agrees with our previous study on multi-fluid STI that shows amplification of TKE to be stronger when there are significant density variations [16, 17].

The density effects on the post-shock PDF of second and third invariants are further explored by comparing the distribution of  $(Q^*, R^*)$ , conditioned on regions with different densities in figure 10 (a)-(c). Figure 10 (a) corresponds to the region of relatively high density ( $\rho > (\bar{\rho} + 90\% \rho'_{rms})$ ), 10 (b) to the region with density around the post-shock mean value, and 10 (c) to the low density region ( $\rho < (\bar{\rho} - 90\% \rho'_{rms})$ ). Before the shock wave, the PDFs corresponding to these regions are also compared and they are all very close to the single-fluid case (not shown). After the shock

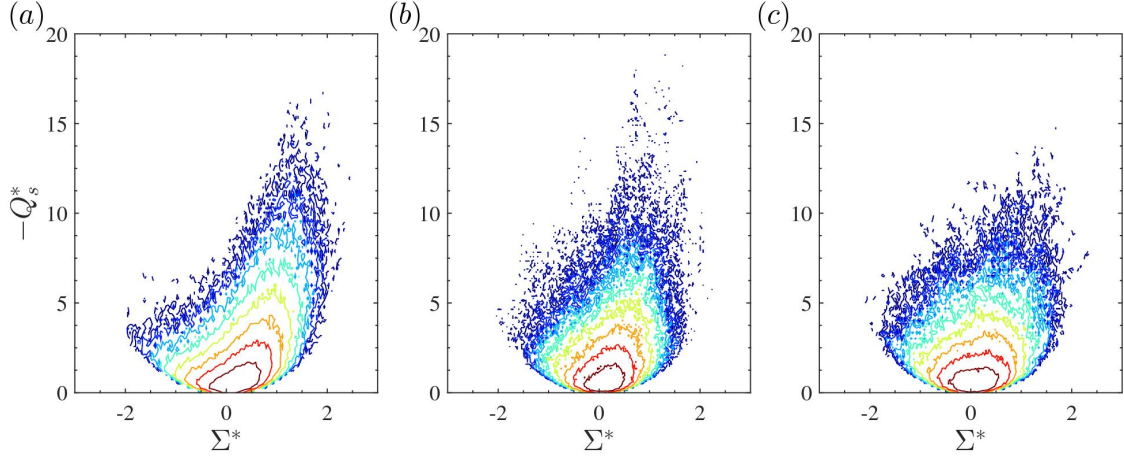


**Fig. 10** Iso-contour lines of post-shock ( $k_0x = 0.44$ ) joint PDF of  $(Q^*, R^*)$ , second and third invariant of and anisotropic part of the velocity gradient tensor in regions with different density values. (a,c) regions with high or low density values ( $\rho > (\bar{\rho} + 90\% \rho'_{rms})$  or  $\rho < (\bar{\rho} - 90\% \rho'_{rms})$ ), (b) regions with density around the post-shock mean value.

wave, the joint PDFs demonstrate significant differences between regions with different densities. Thus, in regions with density closer to that of post-shock mean density, the distribution of invariants appears to be very similar to that shown in figure 9. But for regions with higher density (figure 10 (a)), two important differences are observed. First, the joint PDF becomes more symmetric compared to those in the medium density region or single-fluid case. There is a much larger portion of data points having a local topology of stable-node/saddle/saddle. Secondly, fewer data points fall into first and second quadrants, indicating strain-dominated regions. On the other hand, post-shock regions with low-density values (figure 10 (c)) exhibit characteristics similar to that of isotropic turbulence. As can be seen in figure 10 (c), the joint PDF of  $Q^*$  and  $R^*$  in low-density regions shows almost the same tear-drop shape, only with a larger variance or a wider distribution. The difference is hypothesized to be related to the stronger shock strength variation in the multi-fluid case. It was observed in the current study that the local shock strength is positively correlated with pre-shock density or post-shock density. With a stronger shock effect, the two-dimensionalization effect on post-shock turbulence is stronger in the high-density region. For low-density regions, the weaker two-dimensionalization effect reduces the trend of symmetrization. Moreover, the relatively lower inertia in these regions could make the return to isotropic turbulence process faster. The different characteristics of  $(Q^*, R^*)$  plane in regions with different densities provide additional evidence for the previous argument on preferential amplification of strain and rotation tensor, depending on density.

In our previous study in [17], the axisymmetric state of multi-fluid post-shock turbulence is explored by considering the normalized turbulent stretching term in the vorticity variance equation. It was shown that the normalized turbulence stretching is close to zero immediately after the shock and the multi-fluid turbulence exhibits a faster increase in turbulent stretching. In this paper, we have shown that the multi-fluid joint PDF of  $(Q^*, R^*)$  becomes more symmetric and explore more on how the vortex stretching rate is affected by the axisymmetric state and different turbulence topology.

Similar to what has been used for  $Q^*$  and  $R^*$ , here we define  $(Q_w^*, R_w^*)$  and  $(Q_s^*, R_s^*)$  as the normalized second and third invariant of the rotation ( $W_{ij}$ ) and strain rate ( $S_{ij}$ ) tensors. The rate at which the vorticity is stretched or contracted, or the normalized vortex stretching rate can be conveniently calculated using the formula:  $\Sigma^* = (R_s^* - R_w^*)/Q_w^*$  [28]. In figure 11, the joint PDF of  $(-Q_s^*, \Sigma^*)$  is plotted to investigate the effects of strain field on the vortex stretching rate. Positive and negative  $\Sigma^*$  corresponds to the vortex being stretched and contracted. Figure 11 (a) shows the joint PDF of  $(-Q_s^*, \Sigma^*)$  for isotropic turbulence. The results agree very well with a previous study [28] that the flow is favored by the positive  $\Sigma^*$  or vortex stretching structure, especially in the strong strain dominated region. Here we compare the results from isotropic turbulence to those from single-fluid and multi-fluid post-shock turbulence to develop a better understanding of the post-shock turbulence structure. We note that in figure 11 (b), the joint PDF becomes more



**Fig. 11** Iso-contour lines of joint PDF of  $(-Q_s^*, \Sigma^*)$  for (a) isotropic turbulence and (b,c) single-fluid and multi-fluid cases at post-shock position  $k_0 x = 0.44$

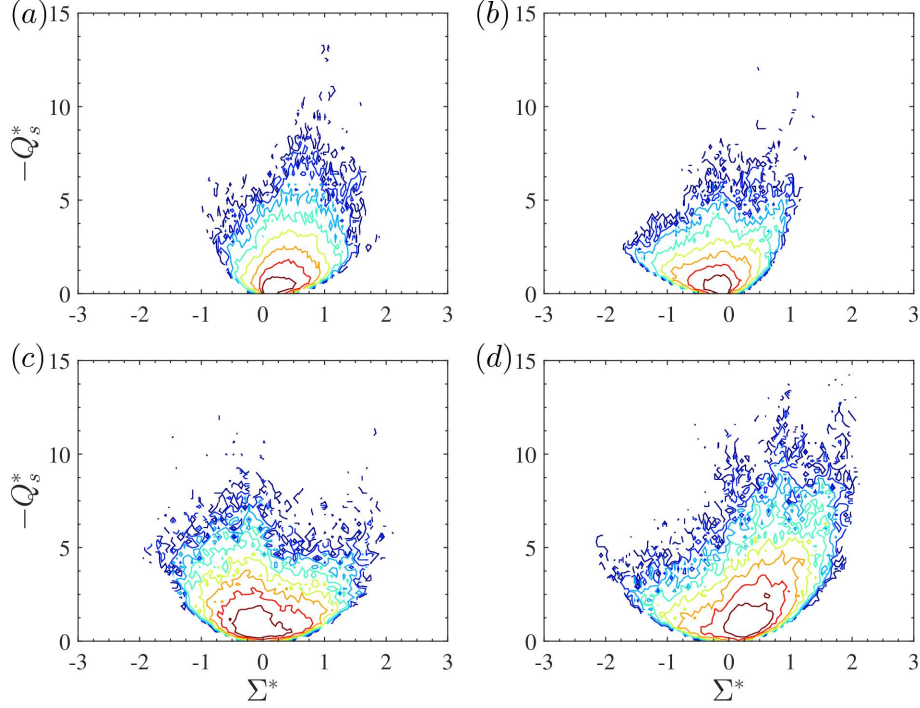
symmetric around  $\Sigma^* = 0.0$  after passing the shock wave, meaning that there are more vortex contraction regions in the post-shock turbulence. Also, the variance of the vortex stretching rate seems to be reduced. Combining these two effects, the averaged vortex stretching rate is reduced after crossing the shock wave. For the multi-fluid case shown in figure 11 (c), the joint PDF becomes almost symmetric, especially in the low  $-Q_s^*$  region. This has a strong influence on low vortex stretching rate for the multi-fluid post-shock turbulence, because the positive and negative  $\Sigma^*$  tend to cancel each other through averaging. Moreover, the variances of the stretching rate between the two cases are almost the same, meaning that the lower stretching rate is mainly due to the rearrangement of the turbulence structure (or more negative  $\Sigma^*$ ), instead of the smaller magnitude of  $\Sigma^*$ .

To understand the contribution from different topology to the vortex stretching rate, the joint PDF of  $(-Q_s^*, \Sigma^*)$  are conditioned on different quadrants for the multi-fluid case. Figure 12 (a,b) shows the joint PDF for regions in the second and first quadrant, or regions with a local topology of stable focus stretching and unstable focus contracting. It can be seen that in these rotation dominated regions, the magnitude of the vortex stretching rate  $\Sigma^*$  is relatively small. Moreover, the first quadrant is dominated by the negative  $\Sigma^*$  and the second quadrant is dominated by positive  $\Sigma^*$ , which can be inferred from their corresponding topology. However, for the third and fourth quadrant, it is shown in figure 12 (c,d) that the magnitude of the vortex stretching rate is larger than the rotation dominated regions (figure 12(a,b)). In the third quadrant, the joint PDF is relatively symmetric and seems to be slightly favored towards the negative vortex stretching rate. The fourth quadrant, on the other hand, is dominated by the positive vortex stretching. We have already observed an increased possibility of multi-fluid post-shock turbulence to fall into first and third quadrants. Considering that the turbulence is dominated by negative vortex stretching in these two quadrants, the lower averaged vortex stretching rate in the multi-fluid case results from increased regions in the first and third quadrants.

## B. Development of Turbulence Structure Following Fluid Particles

In the previous section, we have shown that in the multi-fluid STI, there is an enhancement of unstable focus contracting and stable-node /saddle /saddle structure (1st and 3rd quadrant), especially in regions with relatively high-density values. These data points can be simply extracted from the Eulerian results. However, in order to also study the transient development of post-shock turbulence, one can examine the dynamics of the turbulent structure by following the trajectory of fluid particles using Lagrangian data. The Lagrangian description of turbulence has been widely used in studying some important aspects of turbulence, such as particle dispersion, clustering, and mixing. Non-inertial particles moving with the local flow velocity can shed light on transient processes, which are difficult to study using Eulerian data [29]. To the authors' knowledge, there has been no study of the history of the flow structures in STI via Lagrangian/particle method as has been done in other canonical turbulence problems (e.g. Ref. [30] for isotropic turbulence). In this study, we would like to show some preliminary results on the development of turbulent structures in a Lagrangian reference of frame, with the hope of providing a basic understanding of the flow dynamics.

For the current simulations, we have tracked more than 1.048 million particles that are initialized uniformly at the

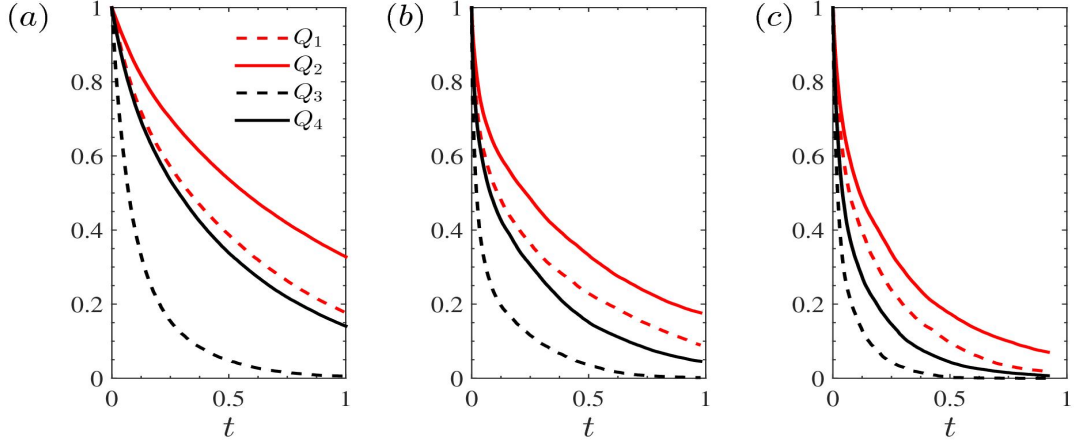


**Fig. 12** Iso-contour lines of joint PDF of  $(-Q_s^*, \Sigma^*)$  for different quadrant immediately after the shock wave. (a)  $Q_2$ , (b)  $Q_1$ , (c)  $Q_3$  and (d)  $Q_4$

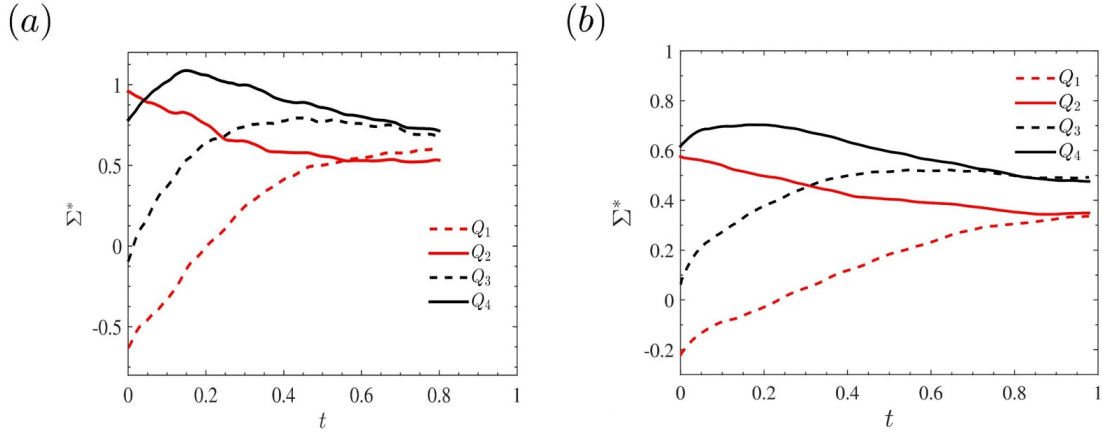
same streamwise position right after the shock wave at  $k_0x = 0.44$ , and calculated the turbulent structures following their trajectories. The aim is to understand the development and evolvement of flow structures following fluid particles and their effects on the turbulent statistics. As shown in figure 1, the particles are initialized like a sheet of particles at the streamwise plane marked by the red lines. These particles are then convected by the turbulent velocity and moved to another plane marked by the blue lines. At this stage, the particle sheet is distorted by the local turbulent structure as shown in figure 1 (b).

The first thing that we consider is the history of particles starting in each of the four quadrants. In figure 1, the percentage of fluid particles that stay in each quadrant are plotted over time so that we can identify the residence time of particles for different flow structures. In figure 13 (a), the percentage of resident fluid particles are plotted for decaying isotropic turbulence as a reference. It is noted that  $Q_3$  and  $Q_4$ , which are the strain dominated regions, have the smallest residence times among all the four quadrants, with  $Q_3$  being the smaller of the two. For the rotation dominated region, the residence time is usually longer, especially for  $Q_2$ . These rotation dominated regions are sometimes identified as coherent structures like stretched vortex tubes. When the particle travels through these structures, they usually stay longer. In figure 13 (b,c), similar results are shown for single-fluid and multi-fluid simulations. Thus, for both cases,  $Q_3$  always has the least residence times and  $Q_2$  has the largest residence times. Comparing all three cases, the particles in multi-fluid and single-fluid post-shock turbulence are shown to evolve faster than in isotropic turbulence. This might be due to the fact that post-shock turbulence experiences a return to isotropy, while the isotropic turbulence already reached an equilibrium among all the quadrants. Between the two post-shock turbulence fields, the multi-fluid case presents a relatively shorter residence times. This agrees with the fact that the multi-fluid turbulence exhibits a faster return to isotropy process than the single-fluid case [17].

In figure 14, the contribution to vortex stretching rate from particles that are initialized in each of the quadrants are plotted following these particles. As expected, at  $t = 0.0$ , particles from  $Q_2$  and  $Q_4$  have positive vortex stretching, while those from  $Q_1$  have negative vortex stretching. This is in good agreement with the joint PDF of  $(-Q_s^*, \Sigma^*)$  as shown in figure 12. For  $Q_3$ , the initial contribution is different between single-fluid and multi-fluid cases, with negative contribution from multi-fluid simulation and positive contribution from single-fluid contribution. As the fluid particles moving with the turbulence, their contributions to the vortex stretching also change. It can be seen that the fast increase



**Fig. 13** Percentage of fluid particles that stay in each quadrant following particles initialized uniformly in (a) isotropic turbulence and (b,c) single-fluid and multi-fluid cases at post-shock position  $k_0 x = 0.44$



**Fig. 14** Contribution to vortex stretching rate from particles in each quadrant following particles initialized uniformly in (a) multi-fluid and (b) single-fluid cases at post-shock position  $k_0 x = 0.44$

in vortex stretching can be mainly attributed to the particles in  $Q_1$  and  $Q_3$ .  $Q_4$  also has an increasing vortex stretching for a short period before its value starts to decrease. The single-fluid case has similar behavior, but the transients process are much weaker. For both cases, averaged vortex stretching rate of  $Q_2$  decreases following the particles.

#### IV. Conclusion

Accurate shock-capturing turbulence-resolving Eulerian simulations, together with a Lagrangian particle tracking algorithm, are used to investigate the interaction of isotropic turbulence with a normal shock wave for both small and large density variation cases. The main objective of this paper is to shed some lights the variable density effects on turbulence structure by comparing multi-fluid STI with the single-fluid case. The numerical accuracy of the algorithm is established by performing both grid and LIA convergence tests. The results show that the turbulence statistics are grid-converged and the LIA prediction can be approached by varying either  $M_t$  or  $Re_\lambda$ , indicating good accuracy of the current numerical method.

Low order statistics are discussed first to show the variable density effects on STI. The multi-fluid case exhibits intensified density variations and TKE after the shock wave. A shorter transient region before the peak is also observed for multi-fluid TKE. Conditional statistics show that there exists a preferential amplification of TKE in regions with relatively low and high density.

The density effects on the post-shock turbulence structure are then studied using Eulerian data. The non-Gaussianity of the velocity derivative is studied by examining the skewness and flatness across and after the shock wave. It is shown that density effects are weak across the shock, but stronger during the post-shock development. The preferential amplification of rotation and strain rate tensors is also affected by density variations and leads to a weaker correlation between the two quantities in the multi-fluid case. This is then shown to be caused by the different roles that density plays on the modification of rotation and strain rate tensor across the shock wave. The variable density effects on turbulent structures are then examined by comparing the joint PDF of the second and third invariants of the velocity gradient tensor. The pre-shock joint PDF has the classic tear-drop shape. However, after the shock wave, a tendency towards symmetrization of the joint PDF, as in single-fluid STI, is observed for the multi-fluid case, with more data points falling into the first and third quadrants. After conditioning the joint PDFs based on fluid density, large differences among heavy, medium, and light fluid regions are observed. In the heavy fluid regions, the joint PDF becomes almost completely symmetrical with an increasing portion of data in the third quadrant. In contrast, the majority of the light fluid data points have a similar distribution to that of isotropic turbulence. A connection between low vortex stretching and the joint  $(Q^*, R^*)$  is established for post-shock turbulence, by considering the contribution to vortex stretching rate from each quadrant.

Lagrangian particles are used to track the development of these turbulent structures. The results show that particles in  $Q_3$  have the least residence time, while those in  $Q_2$  have the longest residence times, within the respective quadrant. Due to the return to isotropy process in the post-shock turbulence, the residence times are smaller than those in isotropic turbulence, especially for the multi-fluid case. It is also shown that particles starting in quadrants  $Q_1$  and  $Q_3$  play an important role in recovering of the vortex stretching rate.

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#### References

- [1] Jammalamadaka, A., Li, Z., and Jaber, F. A., "Subgrid-scale models for large-eddy simulations of shock-boundary-layer interactions," *AIAA journal*, Vol. 51, No. 5, 2013, pp. 1174–1188.
- [2] Jammalamadaka, A., Li, Z., and Jaber, F., "Numerical investigations of shock wave interactions with a supersonic turbulent boundary layer," *Phys. Fluids*, Vol. 26, No. 5, 2014, p. 056101.
- [3] Jammalamadaka, A., and Jaber, F., "Subgrid-scale turbulence in shock-boundary layer flows," *Thoe. Comput. Fluid Dyn.*, Vol. 29, No. 1-2, 2015, pp. 29–54.

- [4] Livescu, D., and Ryu, J., "Vorticity dynamics after the shock–turbulence interaction," *Shock Waves*, Vol. 26, No. 3, 2015, pp. 241–251.
- [5] Hannappel, R., and Friedrich, R., "Direct numerical simulation of a Mach 2 shock interacting with isotropic turbulence," *Appl. Sci. Res.*, Vol. 54, No. 3, 1995, pp. 205–221.
- [6] Mahesh, K., Lee, S., Lele, S. K., and Moin, P., "The interaction of an isotropic field of acoustic waves with a shock wave," *J. Fluid Mech.*, Vol. 300, 1995, pp. 383–407.
- [7] Ryu, J., and Livescu, D., "Turbulence structure behind the shock in canonical shock-vortical turbulence interaction," *J. Fluid Mech.*, Vol. 756, 2014, p. R1.
- [8] Mahesh, K., Lele, S. K., and Moin, P., "The influence of entropy fluctuations on the interaction of turbulence with a shock wave," *J. Fluid Mech.*, Vol. 334, 1997, pp. 353–379. doi:10.1017/S0022112097004576.
- [9] Larsson, J., Bermejo-Moreno, I., and Lele, S. K., "Reynolds- and Mach-number effects in canonical shock-turbulence interaction," *J. Fluid Mech.*, Vol. 717, 2013, pp. 293–321. doi:10.1017/jfm.2012.573.
- [10] Lee, S., Lele, S. K., and Moin, P., "Interaction of isotropic turbulence with shock waves: effect of shock strength," *J. Fluid Mech.*, Vol. 340, 1997, pp. 225–247. doi:10.1017/S0022112097005107.
- [11] Jamme, S., Cazalbou, J.-B., Torres, F., and Chassaing, P., "Direct numerical simulation of the interaction between a shock wave and various types of isotropic turbulence," *Flow Turbul. Combust.*, Vol. 68, No. 3, 2002, pp. 227–268.
- [12] Lee, S., Lele, S. K., and Moin, P., "Direct numerical simulation of isotropic turbulence interacting with a weak shock wave," *J. Fluid Mech.*, Vol. 251, 1993, pp. 533–562. doi:10.1017/S0022112093003519.
- [13] Larsson, J., and Lele, S. K., "Direct numerical simulation of canonical shock/turbulence interaction," *Phys. Fluids*, Vol. 21, No. 12, 2009, p. 126101. doi:10.1063/1.3275856.
- [14] Ribner, H. S., "Convection of a pattern of vorticity through a shock wave," *NACA TR-1164*, 1954.
- [15] Quadros, R., Sinha, K., and Larsson, J., "Turbulent energy flux generated by shock/homogeneous-turbulence interaction," *J. Fluid Mech.*, Vol. 796, 2016, pp. 113–157.
- [16] Tian, Y., Jaber, F., Livescu, D., and Li, Z., "Numerical simulation of multi-fluid shock-turbulence interaction," *AIP Conference Proceedings*, Vol. 1793, AIP Publishing, 2017, p. 150010.
- [17] Tian, Y., Jaber, F., Li, Z., and Livescu, D., "Numerical Study of Variable Density Turbulence Interaction with a Normal Shock Wave," *J. Fluid Mech.*, Vol. 756, 2017, p. R1.
- [18] Li, Z., and Jaber, F. A., "A high-order finite difference method for numerical simulations of supersonic turbulent flows," *Int. J. Numer. Meth. Fl.*, Vol. 68, No. 6, 2012, pp. 740–766. doi:10.1002/fld.2531.
- [19] Lele, S. K., "Compact finite difference schemes with spectral-like resolution," *J. Comput. Phys.*, Vol. 103, No. 1, 1992, pp. 16–42. doi:10.1016/0021-9991(92)90324-R.
- [20] Yeung, P. K., and Pope, S. B., "An algorithm for tracking fluid particles in numerical simulations of homogeneous turbulence," *J. Comput. Phys.*, Vol. 79, No. 2, 1988, pp. 373–416.
- [21] Ristorcelli, J. R., and Blaisdell, G. A., "Consistent initial conditions for the DNS of compressible turbulence," *Phys. Fluids*, Vol. 9, No. 1, 1997, pp. 4–6.
- [22] Lee, S., Lele, S. K., and Moin, P., "Simulation of spatially evolving turbulence and the applicability of Taylor's hypothesis in compressible flow," *Phys. Fluids A: Fluid Dynamics*, Vol. 4, No. 7, 1992, pp. 1521–1530.
- [23] Livescu, D., and Ristorcelli, J. R., "Buoyancy-driven variable-density turbulence," *J. Fluid Mech.*, Vol. 591, 2007, pp. 43–71.
- [24] Livescu, D., "Numerical simulations of two-fluid turbulent mixing at large density ratios and applications to the Rayleigh-Taylor instability," *Phil. Trans. R. Soc. A*, Vol. 371, 2013, p. 20120185.
- [25] Jaber, F. A., Livescu, D., and Madnia, C. K., "Characteristics of chemically reacting compressible homogeneous turbulence," *Phys. Fluids*, Vol. 12, No. 5, 2000, pp. 1189–1209.
- [26] Pirozzoli, S., and Grasso, F., "Direct numerical simulations of isotropic compressible turbulence: Influence of compressibility on dynamics and structures," *Phys. Fluids*, Vol. 16, No. 12, 2004, pp. 4386–4407.

- [27] Wang, J., Shi, Y., Wang, L.-P., Xiao, Z., He, X. T., and Chen, S., “Effect of compressibility on the small-scale structures in isotropic turbulence,” *J. Fluid Mech.*, Vol. 713, 2012, pp. 588–631.
- [28] Ooi, A., Martin, J., Soria, J., and Chong, M. S., “A study of the evolution and characteristics of the invariants of the velocity-gradient tensor in isotropic turbulence,” *J. Fluid Mech.*, Vol. 381, 1999, p. 141–174.
- [29] Yeung, P. K., “Lagrangian investigations of turbulence,” *Annu. Rev. Fluid Mech.*, Vol. 34, No. 1, 2002, pp. 115–142.
- [30] Nomura, K. K., and Post, G. K., “The structure and dynamics of vorticity and rate of strain in incompressible homogeneous turbulence,” *J. Fluid Mech.*, Vol. 377, 1998, pp. 65–97.